

Comprehensive Spatial Stream Pipeline for Hyperspectral Image Analysis

What, How, and Why Explained with Mathematical and Visual Clarity

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Abstract

This document presents an in-depth explanation of a spatial stream deep learning pipeline tailored for hyperspectral image (HSI) analysis. We describe every component in detail, including the intuition, theoretical basis, mathematical formalization, and visual diagrams. This comprehensive report is designed to be both a rigorous academic reference and an accessible educational guide for researchers working on hyperspectral image processing using advanced deep learning architectures.

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1 Introduction and Motivation

Hyperspectral imaging captures spatial images with hundreds of spectral bands, creating rich data cubes essential for detailed tissue analysis in medical imaging and many other applications. The complexity demands a model that captures spectral and spatial context effectively [1, 9].

Intuition: Why Model Both Spatial and Spectral Information?

Combining spatial and spectral information is crucial in hyperspectral imaging because each material interacts uniquely with light across wavelengths, leading to distinct reflectance spectra. These spectra, or spectral fingerprints, help differentiate materials that look similar in RGB images. Spatial context enhances this by capturing how pixel positions relate to one another, revealing continuous structures like tissues and veins. This integration of spectral continuity and spatial coherence improves classification and segmentation, enhancing the accuracy of hyperspectral medical imaging [12].

This leads us to design a hybrid spatial stream pipeline that **jointly models spectral continuity and spatial context**.

2 Hyperspectral Image Data and Challenges

- **Hyperspectral Cube:** $X \in \mathbb{R}^{H \times W \times B}$ (height $\times$ width $\times$ spectral bands).
- **Challenges:**
 - High spectral dimensionality (B can be > 100) leading to Hughes phenomenon [10].
 - Strong spectral correlation leading to redundancy between adjacent bands.
 - Need to preserve fine spatial details (edges, veins, boundaries).
 - Computational and memory burdens for real-time processing.
 - Limited labeled training data for supervised learning [11].

3 Pipeline Overview

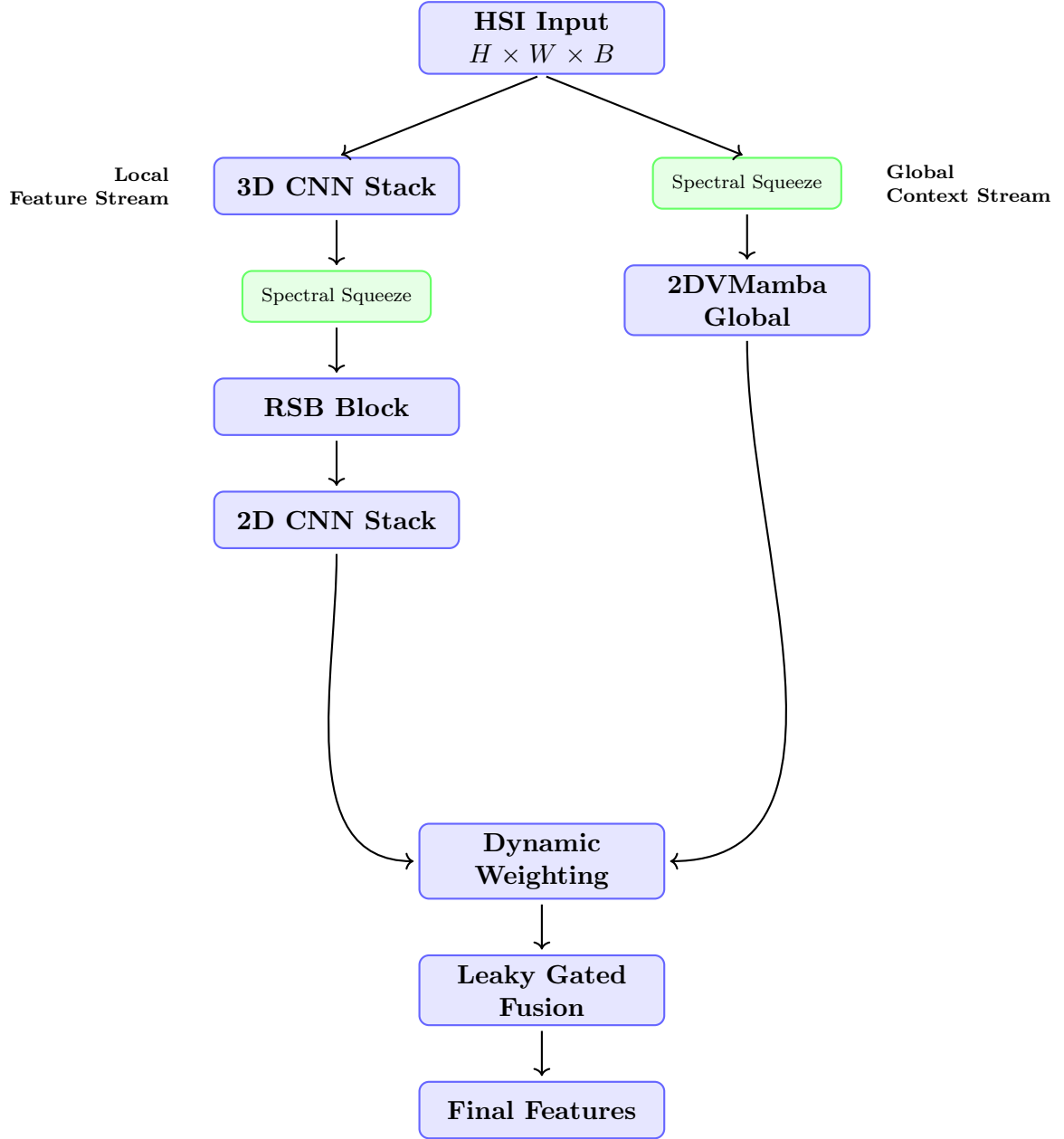


Figure 1: Pipeline overview: local feature extraction (left) and global context modeling (right), merged via dynamic weighting and leaky gated fusion into final features.

4 Detailed Pipeline Architecture

4.1 Local Feature Stream

4.1.1 3D CNN Stack (Spectral–Spatial Extraction)

What: Extract local spectral-spatial features while compressing the spectral bands progressively through joint convolution operations.

How: Three 3D convolution blocks with batch normalization and ReLU activation:

$$\mathbf{F}^{(l)} = \text{ReLU}(\text{BN}(\text{Conv3D}_{K_l}(\mathbf{F}^{(l-1)}))), \quad l = 1, 2, 3$$

where kernel sizes gradually expand along the spectral axis:

$$3 \times 3 \times 3 \rightarrow 3 \times 3 \times 5 \rightarrow 3 \times 3 \times 7,$$

with stride = 1, padding = ‘same’, and channel progression: $C_0 \rightarrow 64 \rightarrow 128 \rightarrow 256$.

Mathematical Formulation: For the l -th 3D convolutional layer:

$$\mathbf{F}_{c,h,w}^{(l)} = \text{ReLU} \left(\sum_{c'=1}^{C_{l-1}} \sum_{k_h=-K_h/2}^{K_h/2} \sum_{k_w=-K_w/2}^{K_w/2} \sum_{k_d=-K_d/2}^{K_d/2} \right. \quad (1)$$

$$\left. \mathbf{W}_{c,c',k_h,k_w,k_d}^{(l)} \cdot \mathbf{F}_{c',h+k_h,w+k_w,d+k_d}^{(l-1)} + b_c^{(l)} \right) \quad (2)$$

- **Spatial dimensions preserved:** Because stride = 1 and padding = ‘same’, height and width remain constant as $H \times W$ throughout the stack.
- **Spectral dimension compressed:** At each block, the depth shrinks ($B \rightarrow B/2 \rightarrow B/4 \rightarrow 1$), reducing redundancy among adjacent spectral bands through learned compression.
- **Adaptive channel expansion:** Channels grow ($64 \rightarrow 128 \rightarrow 256$) in proportion to reduced spectral depth, ensuring enough capacity to capture richer spatial patterns.

Spectral Squeeze Operation: After the final 3D convolution, we perform explicit spectral dimension collapse:

$$\mathbf{F}_{\text{squeeze}} = \text{Squeeze}(\mathbf{F}^{(3)}) \in \mathbb{R}^{H \times W \times 256}$$

where $\text{Squeeze}(\cdot)$ removes the singleton spectral dimension, transforming from $H \times W \times 1 \times 256$ to $H \times W \times 256$.

Output: After squeezing the spectral dimension, the local feature representation is:

$$\mathbf{F}_{C3} \in \mathbb{R}^{H \times W \times 256}.$$

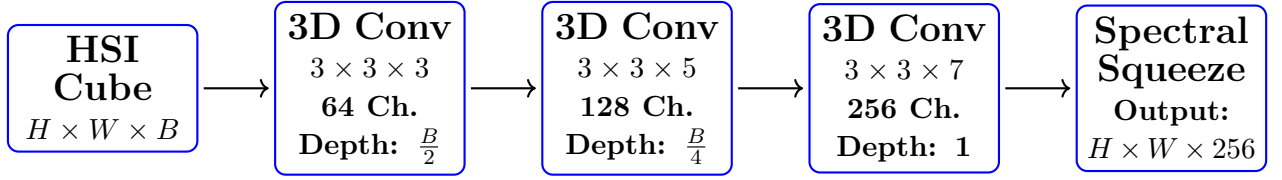


Figure 2: Local feature stream with progressively deeper 3D convolutions compressing spectral dimension and expanding channels.

Intuition: Why this 3D CNN design?

3D convolutions jointly capture local spatial and spectral neighborhoods, which are essential for tissue characterization [1]. The design choices are carefully balanced:

- Larger spectral kernels ($3 \rightarrow 5 \rightarrow 7$) capture broader inter-band correlations and spectral signatures.
- Adaptive channeling ($64 \rightarrow 128 \rightarrow 256$) ensures that as spectral information is compressed, the model increases its feature capacity.
- The constant $H \times W$ resolution preserves fine-grained spatial detail, so that boundary and texture information is never lost during spectral squeeze.

The final tensor $\mathbf{F}_{loc}$ is spectrally compact but spatially expressive, optimized for downstream processing.

4.1.2 Residual Spatial Block (RSB, Spatial Refinement)

Purpose: Acts as a transition layer between 3D CNN (spectral-spatial) and 2D CNN (spatial-only), refining features, aligning channels, and stabilizing training through residual learning [6].

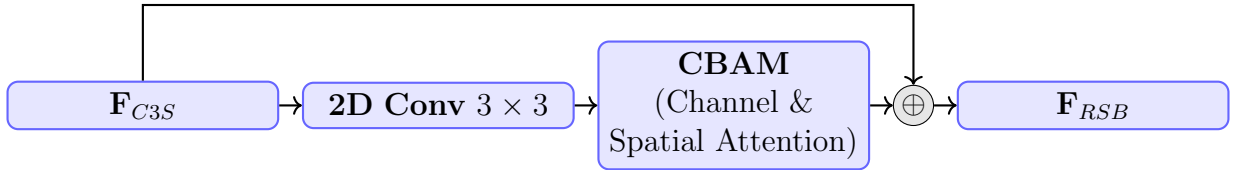


Figure 3: Residual Spatial Block (RSB) with 3×3 convolution and CBAM attention. The skip connection preserves original features, while CBAM adaptively refines them.

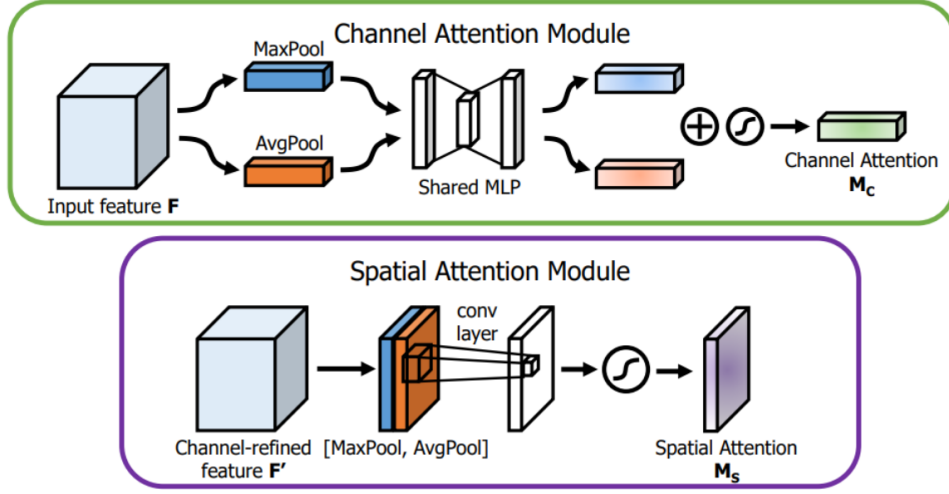
Operation (RSB input):

$$\mathbf{F}_{C3S} \in \mathbb{R}^{C \times H \times W}$$

1. Initial Convolution

$$\mathbf{F}_{conv} = \text{ReLU}(\text{BN}(\text{Conv2D}_{3 \times 3}(\mathbf{F}_{C3S}))), \quad \mathbf{F}_{conv} \in \mathbb{R}^{C \times H \times W}$$

2. CBAM Refinement [2] (Sequential: Channel $\rightarrow$ Spatial)



Channel Attention Module (CAM): The channel attention focuses on 'what' is meaningful given an input image by exploiting inter-channel relationships:

$$\mathbf{F}_{\text{avg}}^c = \text{GAP}(\mathbf{F}_{\text{conv}}) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{F}_{\text{conv}}(i, j) \quad (3)$$

$$\mathbf{F}_{\text{max}}^c = \text{GMP}(\mathbf{F}_{\text{conv}}) = \max_{1 \leq i \leq H, 1 \leq j \leq W} \mathbf{F}_{\text{conv}}(i, j) \quad (4)$$

$$\mathbf{M}_c(\mathbf{F}_{\text{conv}}) = \sigma\left(\mathbf{W}_1(\text{ReLU}(\mathbf{W}_0(\mathbf{F}_{\text{avg}}^c))) + \mathbf{W}_1(\text{ReLU}(\mathbf{W}_0(\mathbf{F}_{\text{max}}^c)))\right) \quad (5)$$

where $\mathbf{W}_0 \in \mathbb{R}^{C/r \times C}$ and $\mathbf{W}_1 \in \mathbb{R}^{C \times C/r}$ with reduction ratio $r = 16$.

$$\mathbf{F}' = \mathbf{M}_c(\mathbf{F}_{\text{conv}}) \odot \mathbf{F}_{\text{conv}}, \quad \mathbf{F}' \in \mathbb{R}^{C \times H \times W}.$$

Spatial Attention Module (SAM): The spatial attention focuses on 'where' is an informative part by utilizing spatial relationships:

$$\mathbf{F}_{\text{avg}}^s = \frac{1}{C} \sum_{c=1}^C \mathbf{F}'_c \quad (6)$$

$$\mathbf{F}_{\text{max}}^s = \max_c \mathbf{F}'_c \quad (7)$$

$$\mathbf{M}_s(\mathbf{F}') = \sigma\left(f^{7 \times 7}([\mathbf{F}_{\text{avg}}^s; \mathbf{F}_{\text{max}}^s])\right) \quad (8)$$

where $f^{7 \times 7}$ represents a 7×7 convolutional layer.

$$\mathbf{F}'' = \mathbf{M}_s(\mathbf{F}') \odot \mathbf{F}', \quad \mathbf{F}'' \in \mathbb{R}^{C \times H \times W}.$$

3. Residual Connection (RSB output)

$$\mathbf{F}_{\text{RSB}} = \mathbf{F}_{\text{C3S}} + \mathbf{F}'', \quad \mathbf{F}_{\text{RSB}} \in \mathbb{R}^{C \times H \times W}.$$

Dimensions

$$\mathbf{M}_c \in \mathbb{R}^{C \times 1 \times 1}, \quad \mathbf{M}_s \in \mathbb{R}^{1 \times H \times W}, \quad \mathbf{F}_{\text{avg}}^c, \mathbf{F}_{\text{max}}^c \in \mathbb{R}^{C \times 1 \times 1}, \quad \mathbf{F}_{\text{avg}}^s, \mathbf{F}_{\text{max}}^s \in \mathbb{R}^{1 \times H \times W}.$$

Intuition: Why RSB with CBAM?

Without RSB, the shift from 3D CNN (spectral-spatial) to 2D CNN (spatial-only) can be abrupt. RSB with CBAM offers:

- **Spatial refinement:** Enhances structures blurred during spectral collapse.
- **Attention mechanism:** CBAM adaptively focuses on important channels and spatial regions [2].
- **Residual learning:** Skip connection aids gradient flow and preserves information [6].
- **Smooth transition:** Stabilizes the 3D→2D handover, reducing feature loss.

RSB is crucial for maintaining feature quality in hybrid 3D–2D pipelines.

4.1.3 2D CNN Stack (Local Spatial Enhancement)

Two 3×3 2D convolution layers with batch normalization and ReLU activation:

$$\mathbf{F}^{(2D1)} = \text{ReLU}(\text{BN}(\text{Conv2D}_{3 \times 3}(\mathbf{F}_{RSB}))) \quad (9)$$

$$\mathbf{F}^{(2D2)} = \text{ReLU}(\text{BN}(\text{Conv2D}_{3 \times 3}(\mathbf{F}^{(2D1)}))) \quad (10)$$

Dilated Convolution Enhancement (Optional): For enhanced receptive field without increased parameters [8]:

$$\mathbf{F}^{(dilated)} = \text{ReLU}(\text{BN}(\text{DilatedConv2D}_{3 \times 3, d=2}(\mathbf{F}^{(2D1)})))$$

where $d = 2$ is the dilation rate, effectively capturing 7×7 receptive field with 3×3 kernel.

Channels are kept at 256 to preserve feature capacity. These layers act as spatial detail refinement, sharpening textures, edges, and local spatial patterns.

Intuition: Why 2D CNN here?

Once spectral information is collapsed, only spatial patterns matter. Small 3×3 filters capture local spatial details efficiently [13]. Optional dilated convolutions expand receptive fields exponentially while maintaining parameter efficiency.

4.2 Global Context Stream: 2DVMamba

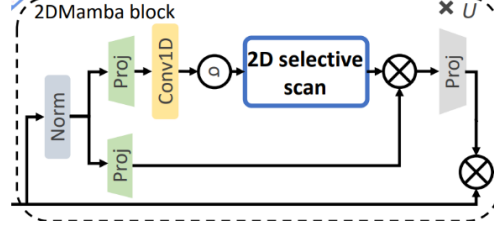
Spectral Squeeze for Global Stream: The input hyperspectral cube is first spectrally collapsed using 1×1 convolution:

$$\mathbf{F}_{glob}^{in} = \text{ReLU}(\text{BN}(\text{Conv1D}_{1 \times 1}(\mathbf{X}))) \in \mathbb{R}^{H \times W \times 256}$$

This operation aggregates spectral information while preserving spatial dimensions for global context modeling.

Design: The spectrally-collapsed features are processed by a single 2DVMamba block:

$$\mathbf{F}_{glob} = \text{2DVMamba}(\mathbf{F}_{glob}^{in}) \in \mathbb{R}^{H \times W \times 256}$$



We use **1 block** (ablation: $1 > 2$), **model width** $C = 256$, and **state dimension** $N = 16$.

Selective State Space Model Formulation [7]: For each state $d = 1..N$ (selective SSM):

$$h_t^d = \tilde{A}_t^d h_{t-1}^d + \tilde{B}_t^d x_t, \quad y_t = \sum_{d=1}^N C_t^d h_t^d$$

with input-conditioned gating:

$$\tilde{A}_t^d = \text{discretize}(A^d, \Delta_t) = \exp(\Delta_t A^d) \quad (11)$$

$$\tilde{B}_t^d = \text{discretize}(B^d, \Delta_t) = (A^d)^{-1}(\exp(\Delta_t A^d) - I)\Delta_t B^d \quad (12)$$

$$C_t^d = C^d(x_t), \quad \Delta_t = \text{softplus}(\Delta(x_t)) \quad (13)$$

2DVMamba Implementation [3]: 2DVMamba applies row-wise then column-wise scans:

$$h_{i,j}^{\text{row}} = \tilde{A}_{i,j} h_{i,j-1}^{\text{row}} + \tilde{B}_{i,j} x_{i,j} \quad (14)$$

$$h_{i,j} = \tilde{A}_{i,j} h_{i-1,j} + \tilde{B}_{i,j} h_{i,j}^{\text{row}} \quad (15)$$

$$y_{i,j} = C_{i,j} h_{i,j} \quad (16)$$

Manhattan distance aggregation:

$$h_{i,j} = \sum_{i' \leq i} \sum_{j' \leq j} \tilde{A}^{(i-i')+(j-j')} \tilde{B} x_{i',j'}, \quad y_{i,j} = C h_{i,j}$$

This formulation shows each output gathers all preceding positions weighted by Manhattan distance, achieving global context modeling while maintaining linear complexity $O(HW)$.

Hardware-Aware Implementation: The 2D selective scan is optimized using:

- **Memory Hierarchy Management:** States kept in SRAM to avoid expensive HBM transfers
- **2D Tiling:** $H \times W$ map split into tiles ($T_H \times T_W$) fitting SRAM
- **Kernel Fusion:** Combined operations reduce memory bandwidth requirements

- **Parallel Processing:** Segmented BlockScan eliminates padding overhead

Intuition: Why 2DVMamba?

2DVMamba achieves global receptive field modeling at **linear complexity** while preserving 2D spatial structure [3]. Unlike Transformers with $O(H^2W^2)$ complexity or flattened 1D Mamba losing spatial relationships, 2DVMamba provides efficient global context modeling essential for hyperspectral analysis.

4.3 Spatial Fusion

Dynamic Feature Weighting [4]

Concatenate local ($\mathbf{F}_{\text{loc}}$) and global ($\mathbf{F}_{\text{glob}}$) features:

$$\mathbf{G} = [\mathbf{F}_{\text{loc}}; \mathbf{F}_{\text{glob}}] \in \mathbb{R}^{H \times W \times 2C}$$

Apply global average pooling and MLP-based weight generation:

$$\mathbf{g} = \text{GAP}(\mathbf{G}) = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \mathbf{G}_{i,j} \in \mathbb{R}^{2C} \quad (17)$$

$$(\alpha, \beta) = \text{Softmax}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{g})), \quad \alpha + \beta = 1 \quad (18)$$

where $\mathbf{W}_1 \in \mathbb{R}^{C/r \times 2C}$ and $\mathbf{W}_2 \in \mathbb{R}^{2 \times C/r}$ with reduction ratio $r = 8$.

Adaptive fusion:

$$\mathbf{F}_{\text{dw}} = \alpha \mathbf{F}_{\text{loc}} + \beta \mathbf{F}_{\text{glob}}$$

Leaky Gated Cross-Attention [5] (Pixel-wise Fusion)

At each pixel (i, j) , we compute cross-attention between adaptive and global features:

$$q_{ij} = \mathbf{W}_Q \mathbf{F}_{\text{dw}}(i, j) \quad (19)$$

$$k_{ij} = \mathbf{W}_K \mathbf{F}_{\text{glob}}(i, j) \quad (20)$$

$$v_{ij} = \mathbf{W}_V \mathbf{F}_{\text{glob}}(i, j) \quad (21)$$

$$s_{ij} = \frac{q_{ij}^\top k_{ij}}{\sqrt{C}}, \quad \gamma_{ij} = \text{softmax}(s_{ij}) \quad (22)$$

Leaky gating with threshold $\tau = 0.1$ and leak factor $\lambda = 0.2$:

$$\tilde{\gamma}_{ij} = \begin{cases} \gamma_{ij}, & \text{if } \gamma_{ij} > \tau \\ \lambda \gamma_{ij}, & \text{otherwise} \end{cases}$$

Final fused feature:

$$\mathbf{F}_{\text{fused}}(i, j) = \tilde{\gamma}_{ij} \mathbf{F}_{\text{dw}}(i, j) + (1 - \tilde{\gamma}_{ij}) v_{ij}$$

Intuition: Why leaky gated cross-attention?

Leaky gating prevents weak attention pixels from being completely suppressed, ensuring training stability [5]. The mechanism adapts fusion based on local feature relevance while maintaining global context flow, crucial for handling varying tissue types in hyperspectral medical imaging.

5 Mathematical Complexity Analysis

Component	Input Shape	Output Shape	Complexity
3D CNN Stack	$H \times W \times B$	$H \times W \times 256$	$O(HWB)$
Spectral Squeeze	$H \times W \times 1 \times 256$	$H \times W \times 256$	$O(1)$
RSB + CBAM	$H \times W \times 256$	$H \times W \times 256$	$O(HW)$
2D CNN Stack	$H \times W \times 256$	$H \times W \times 256$	$O(HW)$
2DVMamba (Global)	$H \times W \times 256$	$H \times W \times 256$	$O(HW)$
Dynamic Fusion	$(H \times W \times 256) \times 2$	$H \times W \times 256$	$O(HW)$
Total			$O(HWB + HW)$

Table 1: Complexity analysis of pipeline components. The overall complexity is dominated by the initial 3D convolutions for spectral processing, while spatial operations scale linearly with spatial dimensions.

6 Implementation Details and Hyperparameters

- **Learning Rate:** Initial LR = 10^{-3} with cosine annealing
- **Batch Size:** 32 for training, 64 for inference
- **Optimizer:** AdamW with weight decay = 10^{-4}
- **Data Augmentation:** Random rotation, flipping, spectral band dropout (0.1)
- **Loss Function:** Cross-entropy with label smoothing ($\epsilon = 0.1$)
- **Regularization:** Dropout (0.1) after fusion, Layer normalization

7 Summary

This comprehensive spatial stream pipeline effectively addresses the key challenges in hyperspectral image analysis through:

- **Dual-stream Architecture:** Local and global feature extraction streams for comprehensive spatial understanding
- **Progressive Spectral Compression:** Systematic reduction of spectral redundancy while preserving essential information
- **Attention Mechanisms:** CBAM for spatial refinement and leaky gated cross-attention for adaptive fusion
- **Linear Complexity:** 2DVMamba enables efficient global context modeling without quadratic scaling
- **Residual Learning:** Skip connections ensure gradient flow and information preservation

The pipeline demonstrates how modern deep learning architectures can be systematically combined to address domain-specific challenges in hyperspectral imaging, providing both theoretical rigor and practical efficiency.

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